Java O-O Program Visualizer/Animator

Milestone 2

By Prottasha Deb, Angelina Martis,
Arya Patel & Aiden Stensrud

Our Program



User Input

Users can input a Java program which is parsed into visualizer instructions.



Visualize

The visualizer will display the input program's memory data.



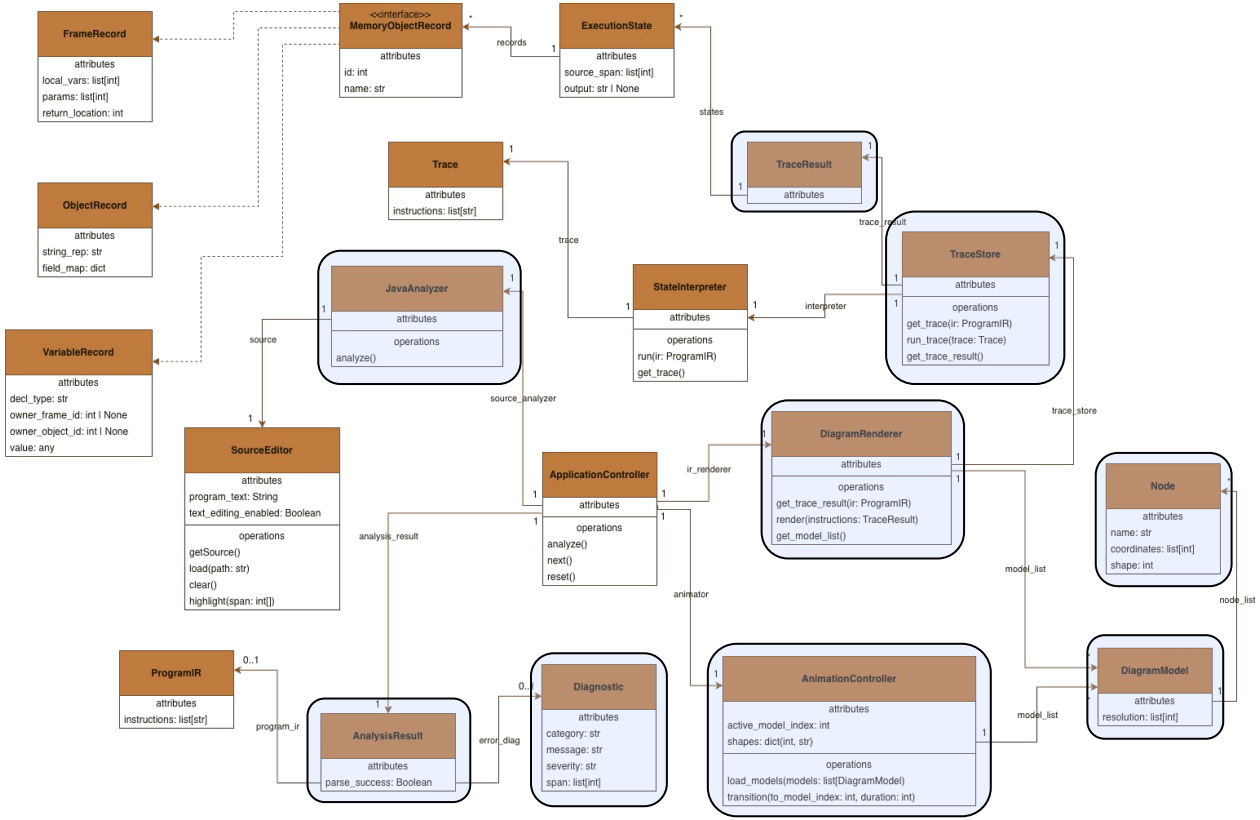
Animate

Users can iterate through the program and see the memory diagram shift.

Work Completed this Milestone

	Prottasha	Angelina	Arya	Aiden
Implement, test, and demo the Java code parser	-	-	-	100%
Implement, test, and demo simple memory diagram generator	-	80%	20%	-
Conceptualize, draft, and demo memory diagram animations	90%	-	-	10%
Resolve technical challenge	-	-	Memory diagram design for complicated memory states	-

Work Completed this Milestone



Java Code Parser



Syntax Analysis

Initial check for invalid character sets such letter/number strings.



Lexical Analysis

Use the language's grammar to check for invalid statements such as unclosed parenthesis.



AST Generation

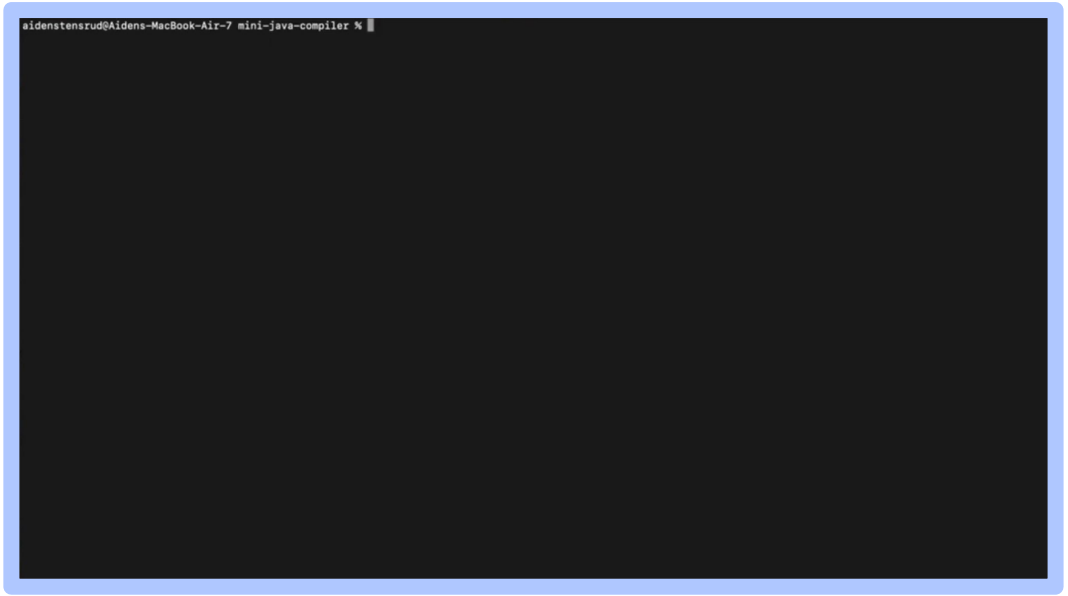
Generate a tree-like data structure from the program and the language's grammar.



Semantic Analysis

Use the AST to check for type mismatches and variable re-declarations.

Java Code Parser Demo



Memory Diagram Animation



Object Visualization

Display Java objects, variables, and their values using memory diagrams.



Reference Animation

Use animated arrows to show references between variables and objects.



Step-by-Step Execution

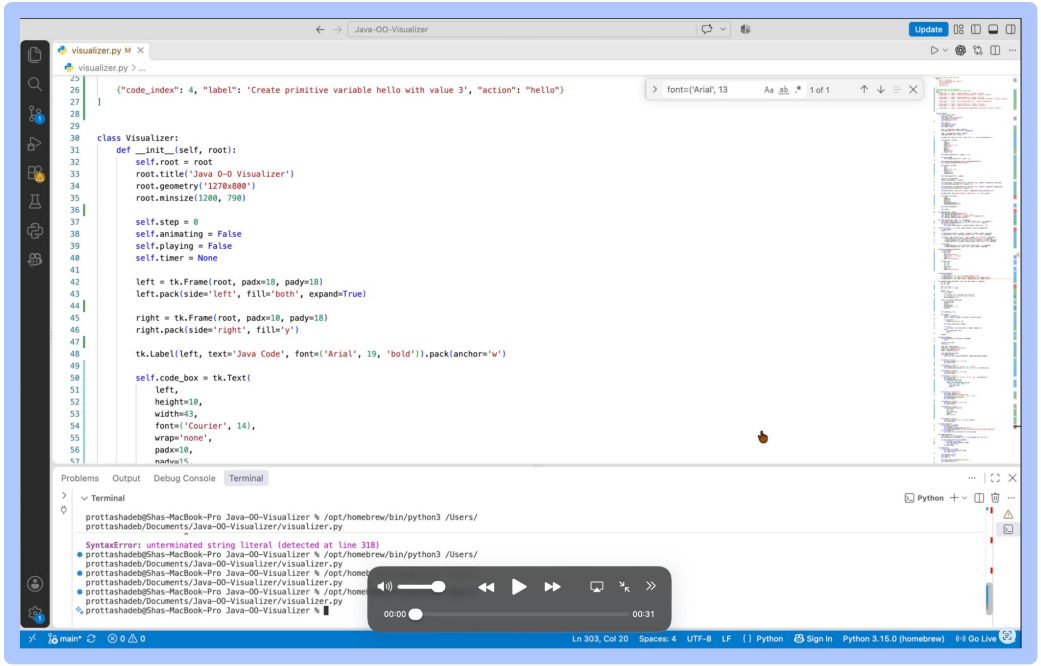
Highlight Java statements and visualize memory changes one step at a time.



Playback Controls

Navigate the simulation using Next Step, Auto Play, and Reset.

Java Code Animator Demo



Memory Diagram Generator



User Input

Users input a text file representing the memory state as a list of variables, class objects, and references.



Generate

The generator draws the variables, objects, and reference pointers according to the information in the text file.



Render

Users can view the rendered memory diagram.

Memory Diagram Generator Demo

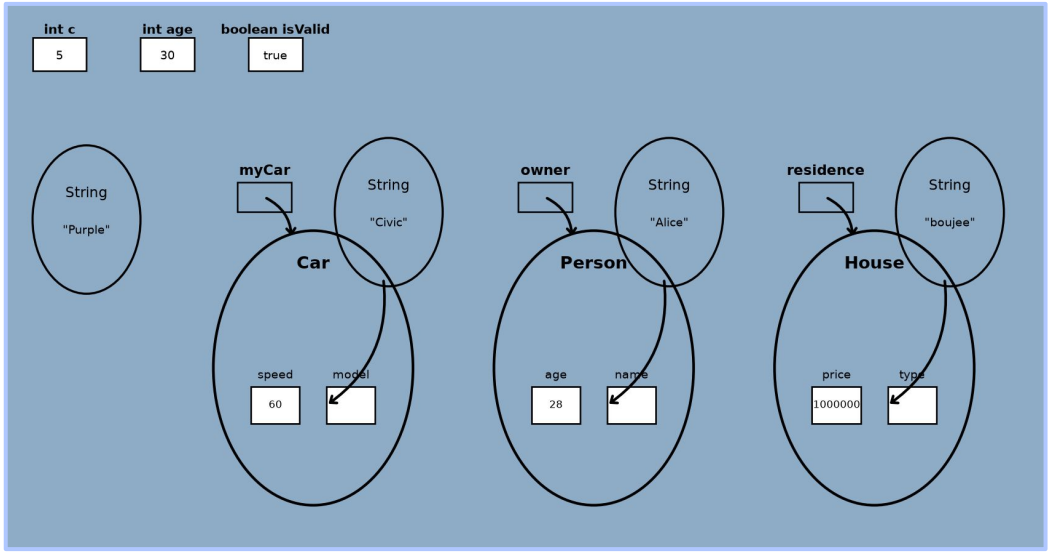
```
# Stack Primitive Variables
variable: int c = 5
variable: int age = 30
variable: boolean isValid = true
object: 0x07 | String | value: Purple

# Stack Reference Variables
variable: myCar -> 0x01
variable: owner -> 0x02
variable: residence -> 0x03

# Objects
object: 0x01 | Car | speed: 60, model: 0x04
object: 0x02 | Person | age: 28, name: 0x05
object: 0x03 | House | price: 1000000, type: 0x06

object: 0x04 | String | value: Civic
object: 0x05 | String | value: Alice
object: 0x06 | String | value: boujee
```

INPUT



OUTPUT

Work for Next Milestone

	Prottasha	Angelina	Arya	Aiden
Implement, test, and demo converter from parsed code to memory representations	-	-	-	100%
Implement, test, and demo memory diagram animator	20%	80%	-	-
Resolve technical challenge	Accurate animations for complex Java programs	-	-	-
Implement, test, and demo GUI	-	20%	70%	10%

Thanks for listening!

Do you have any questions?